EVERY OBJECT OF THE DERIVED CATEGORY OF AN INVERSE SYSTEM OF RINGS IS REPRESENTED BY A SYSTEM OF COMPLEXES WITH SURJECTIVE TRANSITION MAPS

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Allowed standard prerequisites (editorial).

- Homotopy equivalences of complexes and their compatibility with an inverse system. used at proof lines 23792-23799. Is the invariance that allows the representative system to be replaced stage by stage.
- Factoring a map of complexes as a homotopy equivalence followed by a termwise split surjection. invoked at proof lines 23807-23810. Supplies both the homotopy equivalence and the surjectivity at each inductive step.

Source reference. Original lemma 0642: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7851–7851. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let (A_n) be an inverse system of rings. Every $K \in D(Mod(\mathbf{N}, (A_n)))$ can be represented by a system of complexes $(M_n^\bullet)$ such that all the transition maps $M_{n+1}^\bullet \to M_n^\bullet$ are surjective.*

Proof. Let K be represented by the system $(K_n^\bullet)$. Set $M_1^\bullet = K_1^\bullet$. Suppose we have constructed surjective maps of complexes $M_n^\bullet \to M_{n-1}^\bullet \to \ldots \to M_1^\bullet$ and homotopy equivalences $\psi_e : K_e^\bullet \to M_e^\bullet$ such that the diagrams

$$\begin{array}{ccc} K_{e+1}^\bullet & \longrightarrow & K_e^\bullet \\ \downarrow & & \downarrow \\ M_{e+1}^\bullet & \longrightarrow & M_e^\bullet \end{array}$$

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commute for all $e < n$. Then we consider the diagram

$$\begin{array}{ccc} K^\bullet_{n+1} & \longrightarrow & K^\bullet_n \\ & & \downarrow \\ & & M^\bullet_n \end{array}$$

By Derived Categories, Lemma 0642 we can factor the composition $K^\bullet_{n+1} \to M^\bullet_n$ as $K^\bullet_{n+1} \to M^\bullet_{n+1} \to M^\bullet_n$ such that the first arrow is a homotopy equivalence and the second a termwise split surjection. The lemma follows from this and induction. $\square$